

PinPoint: Synchronization-Aware and Explainable Audio–Visual Deepfake Detection

Palak Parmar^a, Chintan Bhatt^{b,*}, SantoshKumar Bharti^a

^a*Department of Computer Science and Engineering, School of Science and Technology,
Pandit Deendayal Energy University, Gandhinagar, Gujarat, India*

^b*University of Wollongong, Australia*

Abstract

Audio–visual deepfakes create a forensic challenge because a detector must identify manipulated media while also exposing evidence that can be inspected by human reviewers. Existing multimodal detectors often fuse acoustic and visual features without explicitly constraining whether the learned interaction reflects temporal synchrony between speech and facial motion. This paper presents PinPoint, a synchronization-aware audio–visual detector designed to make cross-modal alignment both predictive and auditable. PinPoint combines an explainability-compatible ResNet-18 visual encoder, a convolutional Gated Recurrent Unit audio encoder, stacked gated cross-attention blocks, a three-stage curriculum, and a multi-component synchronization loss that encourages aligned real samples to produce diagonal, concentrated, and temporally smooth audio-to-video attention maps. On a unified benchmark constructed from LAV-DF and FakeAVCeleb, PinPoint achieves 97.47% accuracy over 26,097 test samples, with macro F1 of 0.9680 and fake-class F1 of 0.9825. Ablation experiments show that removing the gate, synchronization loss, or curriculum reduces accuracy to 58.35%, 75.47%, and 81.44%, respectively. A multimodal explainable artificial intelligence analysis links the predictions to mouth-region, acoustic-feature, and lip-sync evidence. These results indicate that synchrony-aware supervision can improve deepfake detection accuracy while making model behavior easier to audit.

*Corresponding author.

Email addresses: 24rcp002@sot.pdpu.ac.in (Palak Parmar), cbhatt@uow.edu.au (Chintan Bhatt), santosh.bharti@sot.pdpu.ac.in (SantoshKumar Bharti)

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Declarations

Competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability. LAV-DF and FakeAVCeleb are publicly described datasets. Derived preprocessed tensors and trained model artifacts can be made available by the authors upon reasonable request, subject to the licenses and access terms of the original datasets.

Code availability. The implementation is available in the project repository during review and can be released with the final manuscript subject to institutional approval.

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